

MC injection and signal recovery

Scope. Exact draws of 1,000, 5,000, 10,000 and 30,000 selected candidates from the supplied smeared 2021 MC distributions are injected into 20 background toy experiments at each available generated mass from 60 through 260 MeV. The 40 MeV sample is excluded. Two prespecified masks compare extraction at the generated mass with extraction around the MC core.

Main finding

The shifted mask reduces signal loss at the largest injection, but does not close the recovery test. At $N = 30,000$, after subtracting the paired zero-injection fit, the mean recovered increment is 0.715-0.895 of N with the pole-centered mask and 0.780-0.938 with the shifted mask over 60-240 MeV. The clean-sideband controls identify signal in GP training bins as a material source of the loss.

Primary, 60-240 MeV	Pole-centered	MC-core-shifted
N = 1,000: recovery range	-1.162 to 4.386	-0.702 to 4.469
N = 5,000: recovery range	0.473 to 1.570	0.600 to 1.633
N = 10,000: recovery range	0.631 to 1.216	0.702 to 1.277
N = 30,000: recovery range	0.685 to 0.983	0.752 to 1.042

The table gives the range of mean fitted/injected yield across the ten masses in the original 60-240 MeV direct-template comparison. It is a descriptive range, not a confidence interval. Pointwise Monte Carlo errors and the separate 260 MeV extension appear in the figures.

Interpretation boundary

This is a conditional test of the supplied all-selected candidate shapes and a pinned background truth. The MC contains broad and displaced components, and v6.1 did not establish a pure truth-associated resonance response or complete production-selection equivalence. Recovery of these templates does not establish physical signal efficiency, a coupling limit, global significance or detector-calibrated interval coverage.

Only 20 toys populate each cell. Pull widths and interval-inclusion fractions are imprecise. Exact- N multinomial signal draws differ from the fitted Poisson signal model; even the known-background pull width need not be one. The template and MC center are fixed: finite-MC and detector-response uncertainties are not propagated.

Saved fit rows: 6,160. Primary nonzero-injection cells: 88, each with 20 fits. Reproducible ledgers: results/toys.csv, summary.csv and paired.csv. Study construction and input identities are recorded in protocol.json and the provenance files.

Toy construction and fit protocol

Signal draws and normalization

Each injection draws exactly N entries from the archived, unit-weight, all-selected MC histogram, retaining categories outside the 36-299.75 MeV analysis support. Within-support histogram draws are mapped to the analysis bins. No additional smearing, translation, truth selection or within-window renormalization is applied. The fitted amplitude is in full-selected events: the template sum in the support and in the extraction mask retains the corresponding fractions. Actual support/window counts and expected fractions are saved per toy.

Background truth and the paired ensemble

The background truth is the pinned nominal 2021 all-data GP prediction from v5.8.2. Poisson background fluctuations generate 20 toys at each mass; the same background draw is reused across injection levels and both methods. Within each level, both methods use the same histogram injection. The zero-injection baseline permits a paired incremental-response diagnostic.

The two extraction masks

The pole-centered method places the fit/training exclusion mask at m_0 . The shifted method places it at a fixed MC-only core center $c(m_0)$, inherited from v6.1 where available. Both use the unchanged half-width 2.25 times the nominal resolution at m_0 . Only the mask moves. The MC template remains at its reconstructed masses, and kernel coordinates remain attached to generated mass m_0 . The extension at 260 MeV uses the archived 250 MeV kernel state and is marked separately.

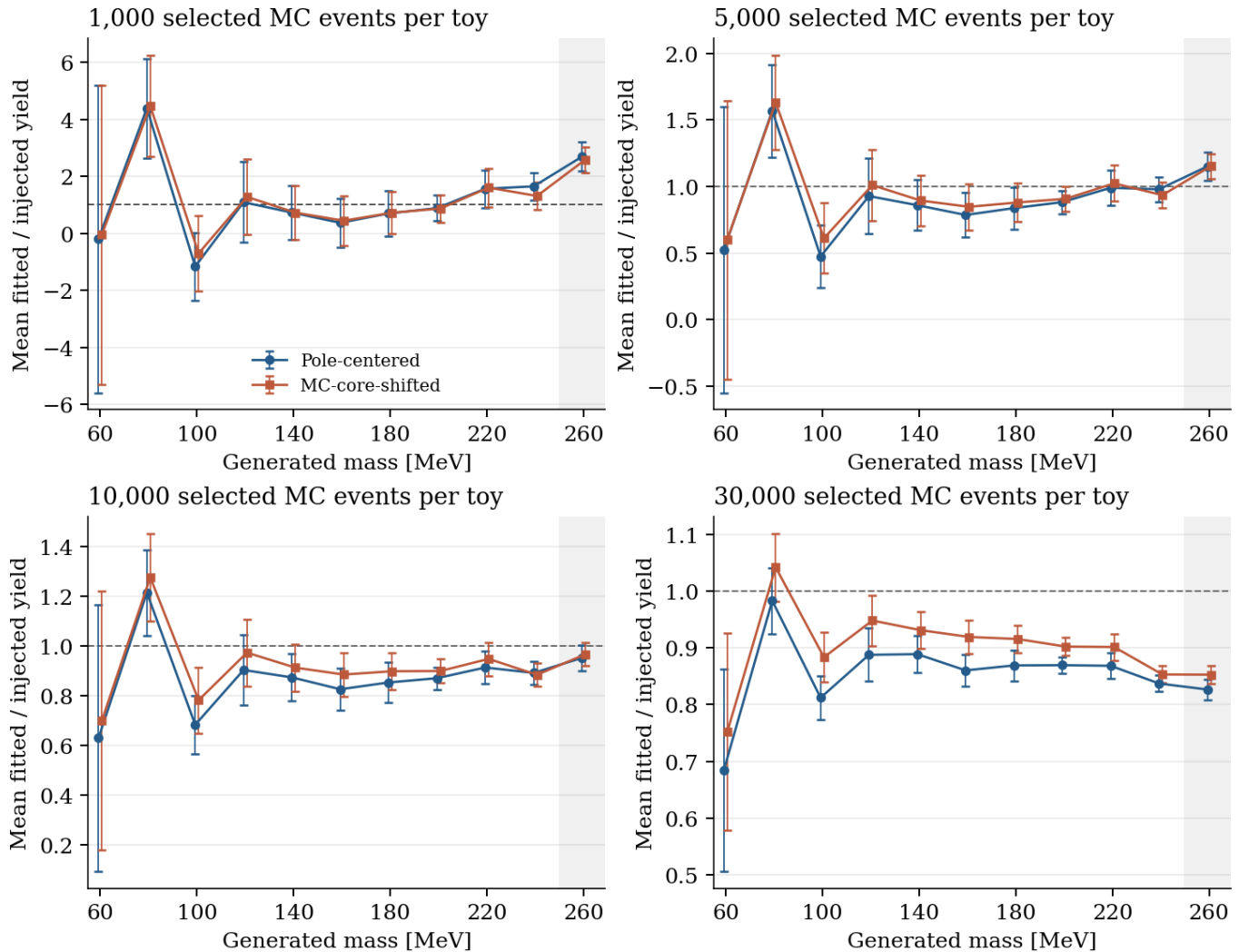
Three prespecified extraction conditions

Injected sidebands are the primary condition: signal entering training bins can affect the GP prediction. Clean sidebands train on the same toy's background-only sidebands while the extraction bins retain the injected signal. The known-background control fixes the background to its generating expectation. In the GP conditions, archived hyperparameters are fixed but the conditional mean and correlated covariance are recomputed for each toy; the likelihood includes the correlated background constraint.

Pulls and interval inclusion

The signed fitted yield A_{hat} is retained for a null diagnostic without a nonnegative-yield boundary. A pull is $(A_{\text{hat}} - N)/\sigma_A$, with σ_A from the profiled likelihood curvature. The profile likelihood ratio at the injected truth gives interval-inclusion indicators for nominal 68.27% (q at most 1) and 95% one-parameter thresholds. Saved 95% Clopper-Pearson intervals quantify the 20-trial sampling uncertainty in those inclusion fractions. These are conditional diagnostic inclusions, not a certification of physical coverage.

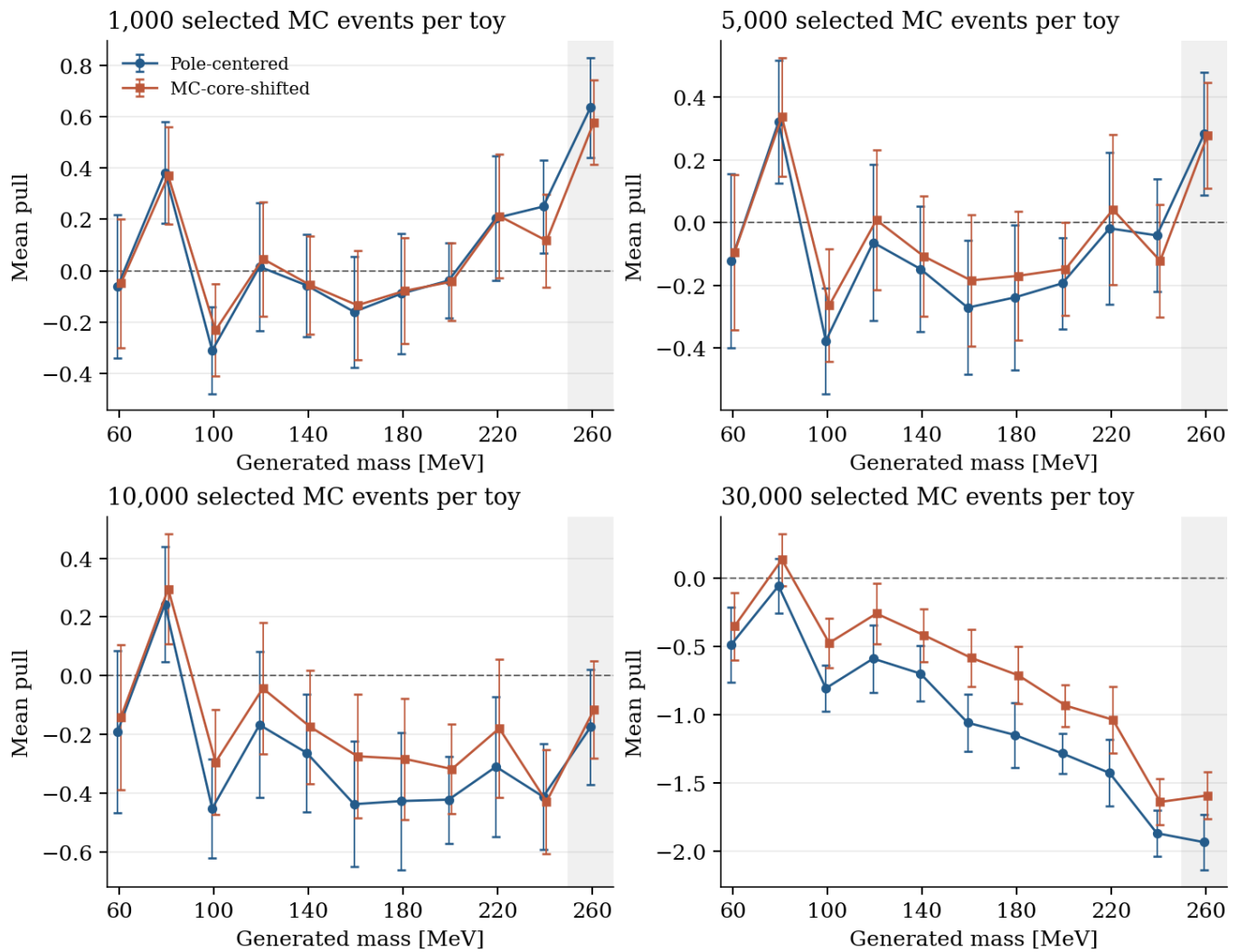
Recovered yield across generated mass



Primary injected-sideband result. Each point is the average fitted full-selected yield divided by the exact injection N . Error bars are the sample standard error across 20 toys. The dashed line denotes exact mean recovery. The gray band marks 260 MeV, outside the earlier 60-240 MeV direct-template comparison and using the archived 250 MeV kernel state.

The mean contains any baseline extraction offset as well as the response to the injected signal. The paired zero-injection subtraction on page 8 separates these effects descriptively. Full-selected yield includes draws outside the analysis support; the template fractions make this normalization explicit.

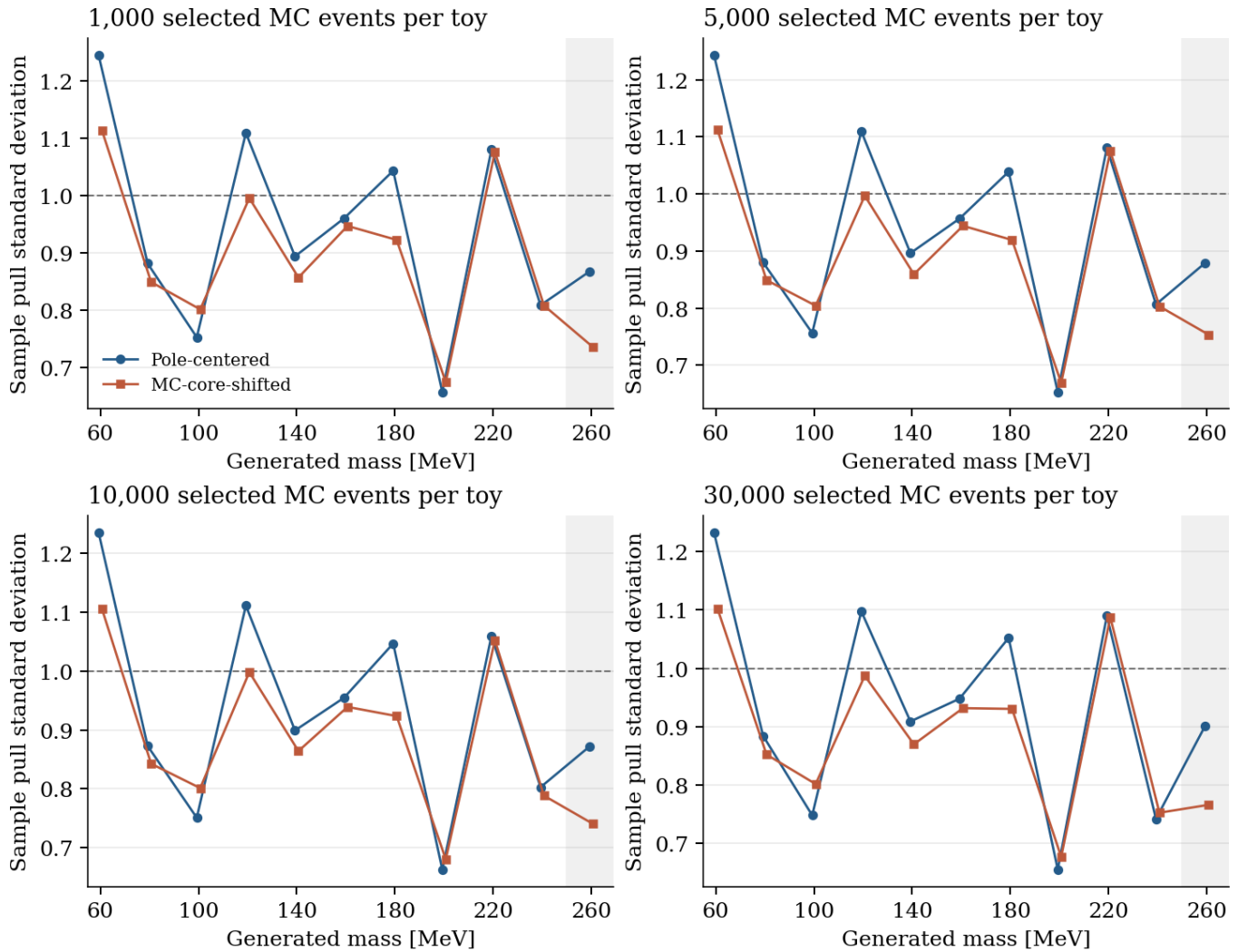
Mean pulls



Each point is the mean of $(\hat{A} - N)/\sigma_A$ in the primary injected-sideband condition. Error bars are the sample pull standard deviation divided by $\sqrt{20}$. A zero mean is the unbiased reference for this conditional ensemble; points sharing toys are correlated.

A near-zero pull mean alone does not show reliable extraction: it must be read with yield recovery, the pull width, the zero-injection control and the fit's quoted uncertainty. With only 20 toys, small deviations are not evidence for a calibrated failure or success probability.

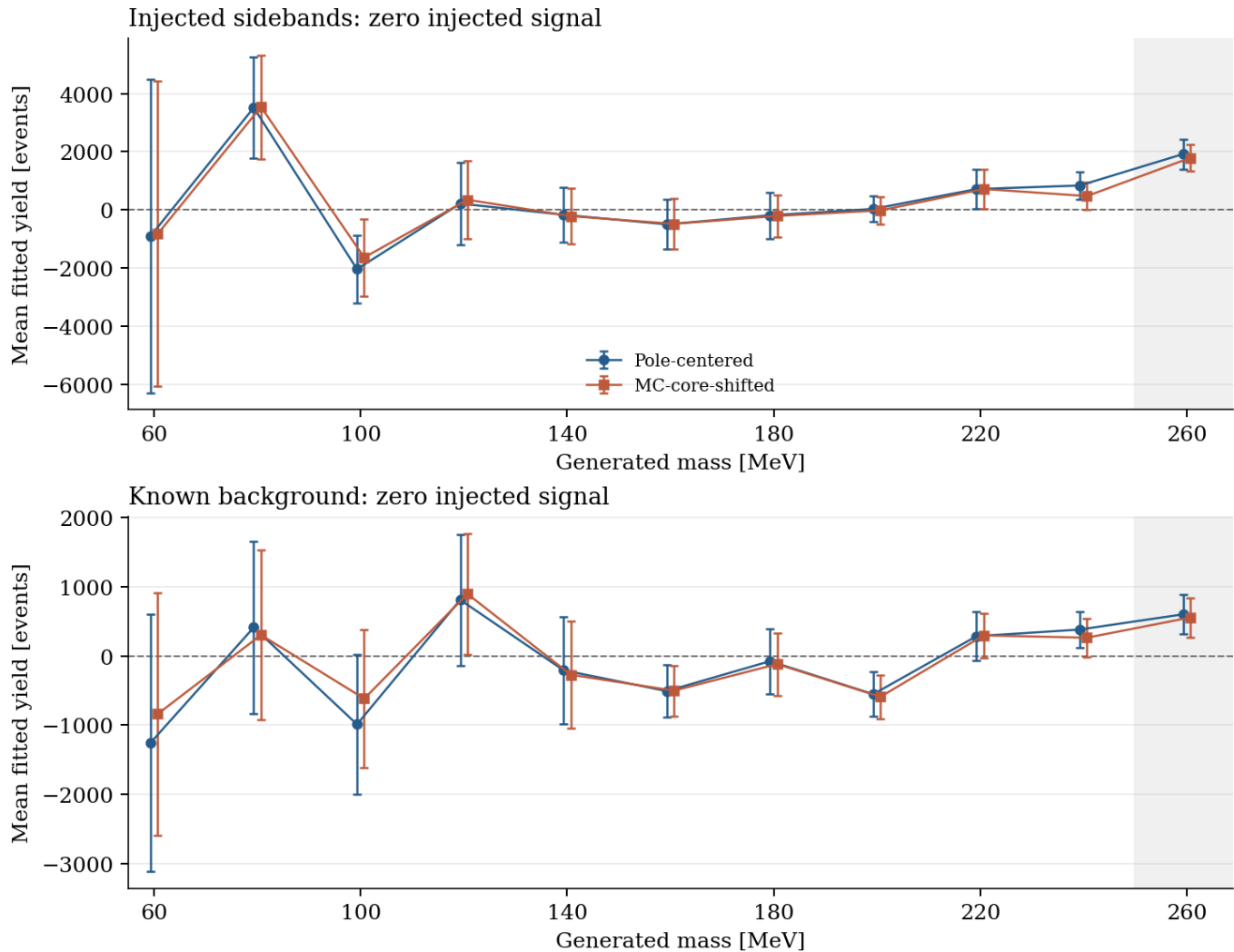
Pull widths



The plotted width is the sample standard deviation of the 20 pulls, using one degree of freedom for the estimated mean. No Gaussian fit to a pull histogram is used. The dashed width-one line is a reference, not an expectation guaranteed by this ensemble.

The injected signal has an exact total N and multinomial fluctuations among bins; the fitted signal likelihood is Poisson. The generating background is fixed conditionally, while GP prediction uncertainty enters the fit. These choices can change widths even in an otherwise consistent estimator. Twenty-toy width estimates are noisy, and the per-mass count galleries preserve their visible distribution shapes.

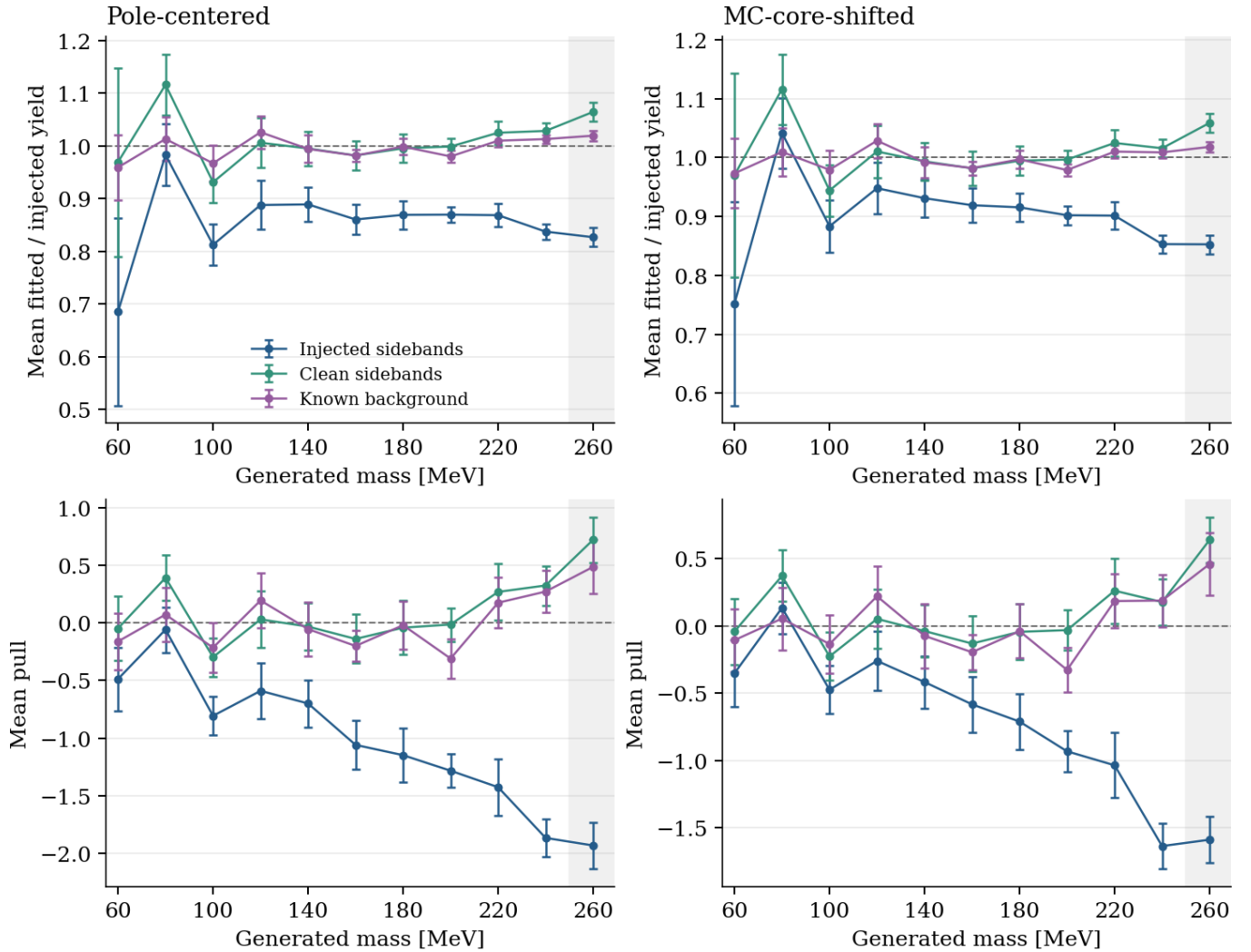
Zero-injection background diagnostic



With $N = 0$, the expected signal yield is zero. Points show the mean signed extracted yield; bars show its sample standard error across the same 20 background toys. The primary and known-background conditions separate background-prediction effects from the fixed-truth likelihood control.

Clean and injected GP sidebands coincide at zero injection, so a duplicate clean-sideband null is unnecessary. A baseline offset can dominate fractional recovery at small injections. The paired incremental response subtracts each toy's corresponding zero-injection fit.

Controls at 30,000 injected events

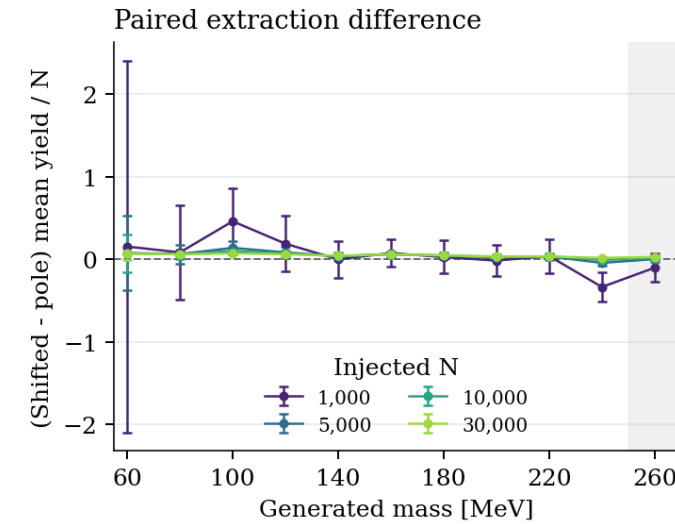


Both masks are shown with three conditions using the same injected toys. Clean sidebands remove the injected component from GP training while preserving it in the extraction bins. Known background fixes the generating expectation and removes GP prediction uncertainty. These are explanatory controls, not candidate production prescriptions.

A deterministic mean-count (Asimov) example at 240 MeV and $N = 30,000$ yields 24,023/24,859 events for injected sidebands, 29,743/29,724 for clean sidebands, and 30,000/30,000 for known background (pole/shifted). The loss thus persists without Poisson toy fluctuations. Known background also removes the GP nuisance constraint, so its comparison is not a single-factor replacement for the GP fit.

Across both masks and 60-240 MeV, mean recovery is 0.931-1.116 with clean sidebands and 0.959-1.029 with known background. These descriptive ranges support the sideband-absorption mechanism while retaining baseline and finite-toy limitations.

Paired mask and incremental response

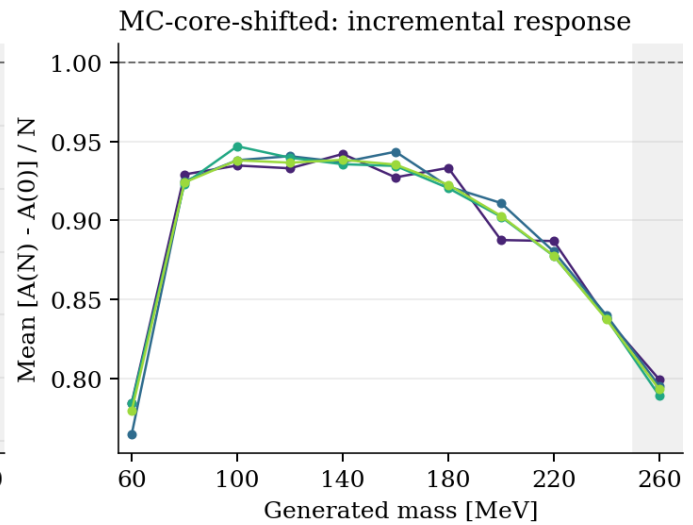
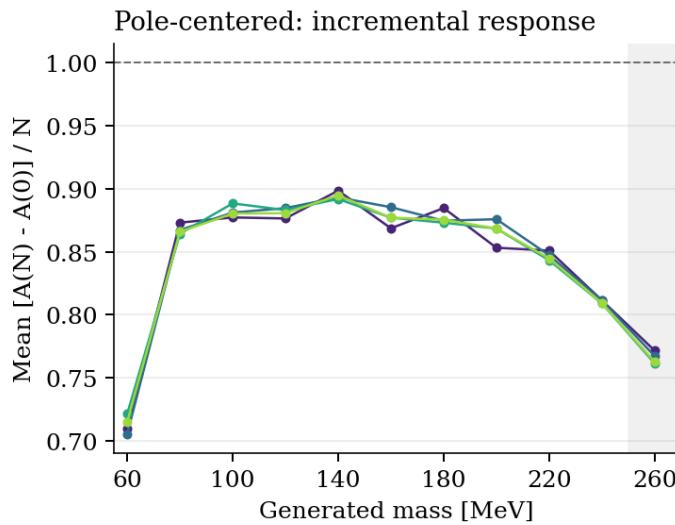


Same toy, two masks

Upper left: a positive value means the shifted mask extracts more signal.

Below: subtract each method's paired zero-injection fit before dividing by N. This removes its toy-specific baseline offset, but does not correct bias.

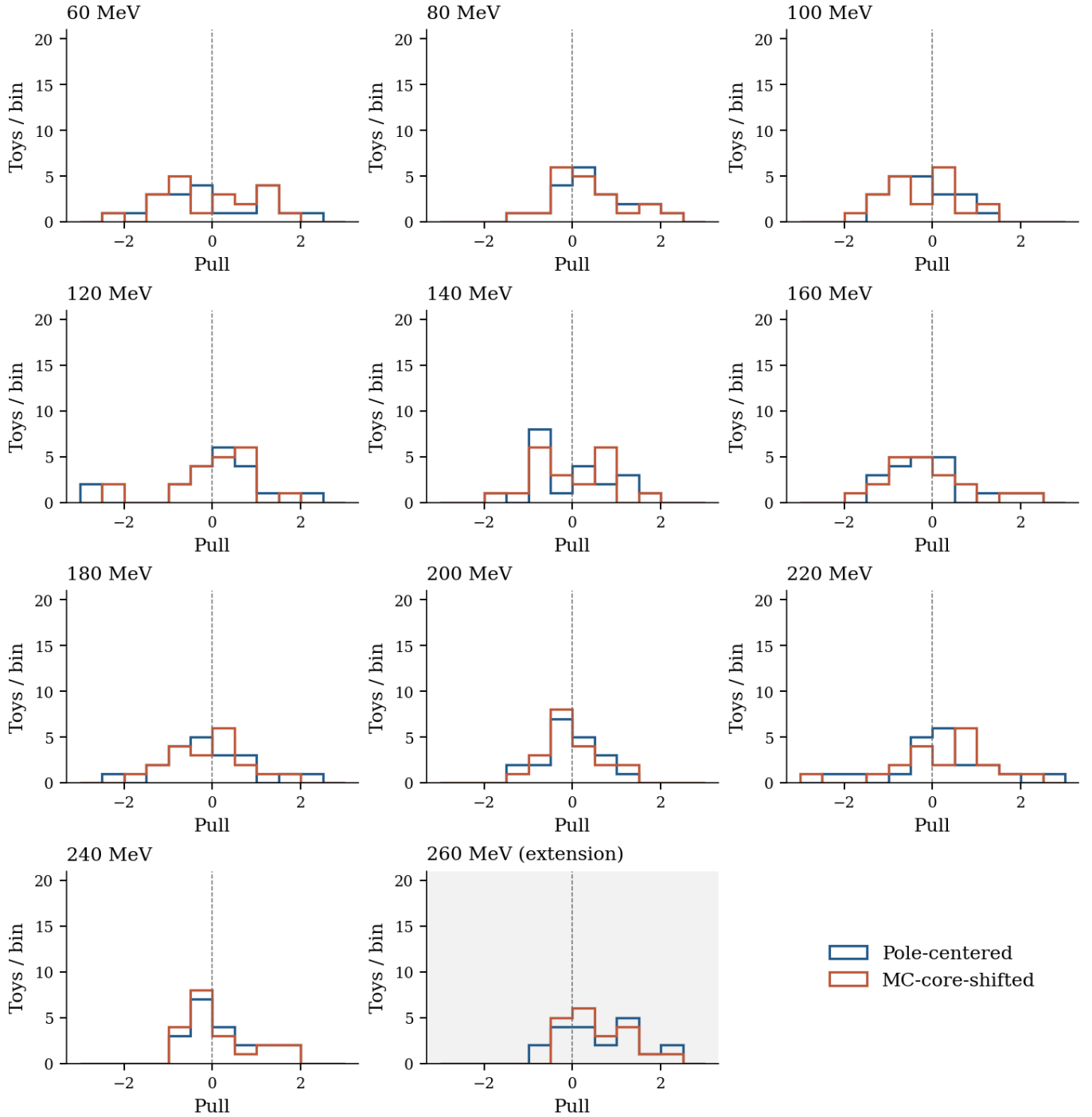
The two estimates share the same background and MC injection draws.



The upper panel compares shifted and pole-centered yields within each identical toy; error bars use the standard error of the paired difference. Lower panels show the mean fitted increment after subtracting that same toy's zero-injection fit. These quantities retain the full-selected normalization.

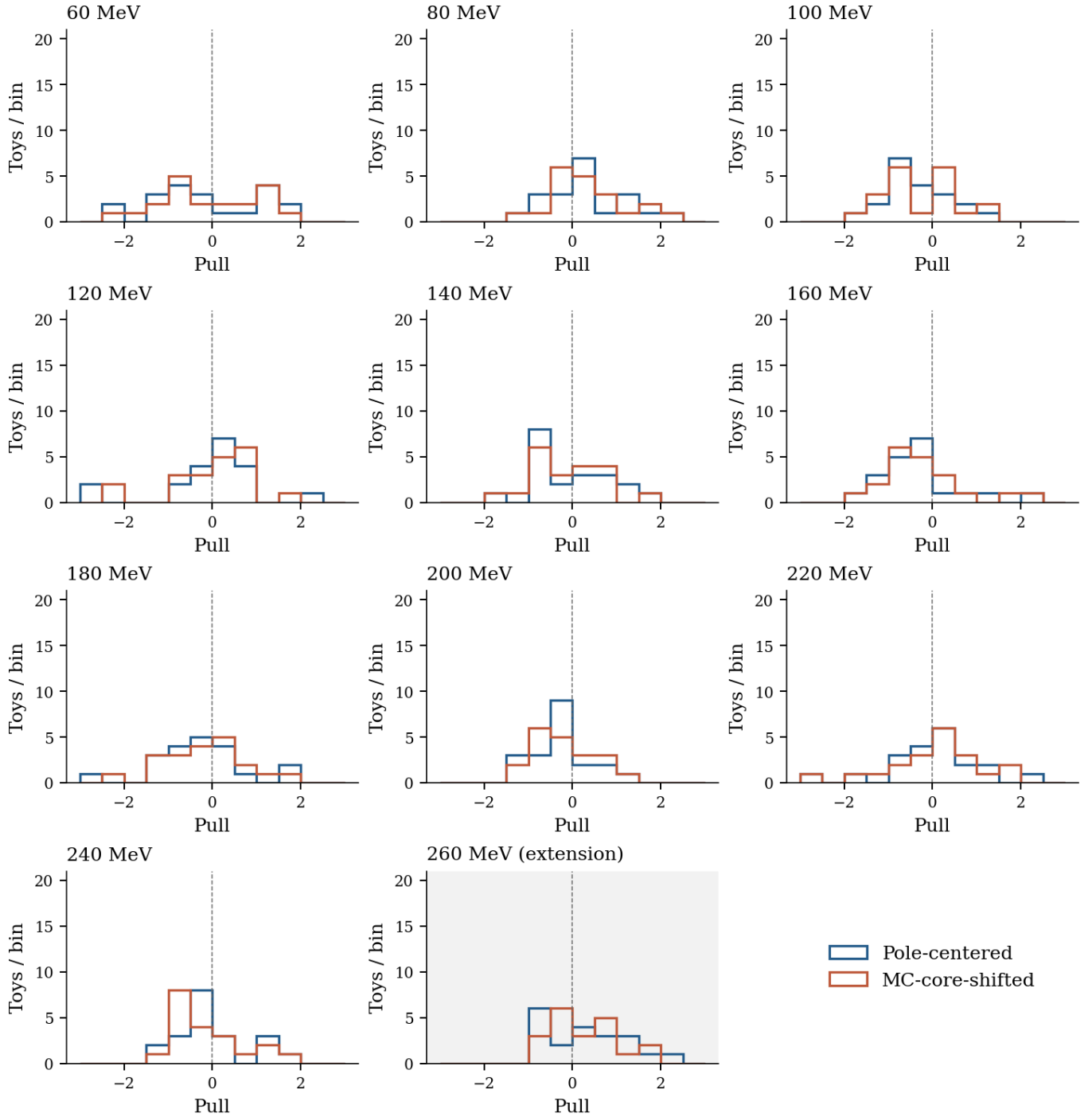
Baseline subtraction is a diagnostic only. It is not applied to the primary estimator and does not remove signal absorbed by sideband training. A larger extracted yield does not by itself establish a better mask; bias, dispersion and interval inclusion must be assessed together.

Pull distributions: N = 1,000



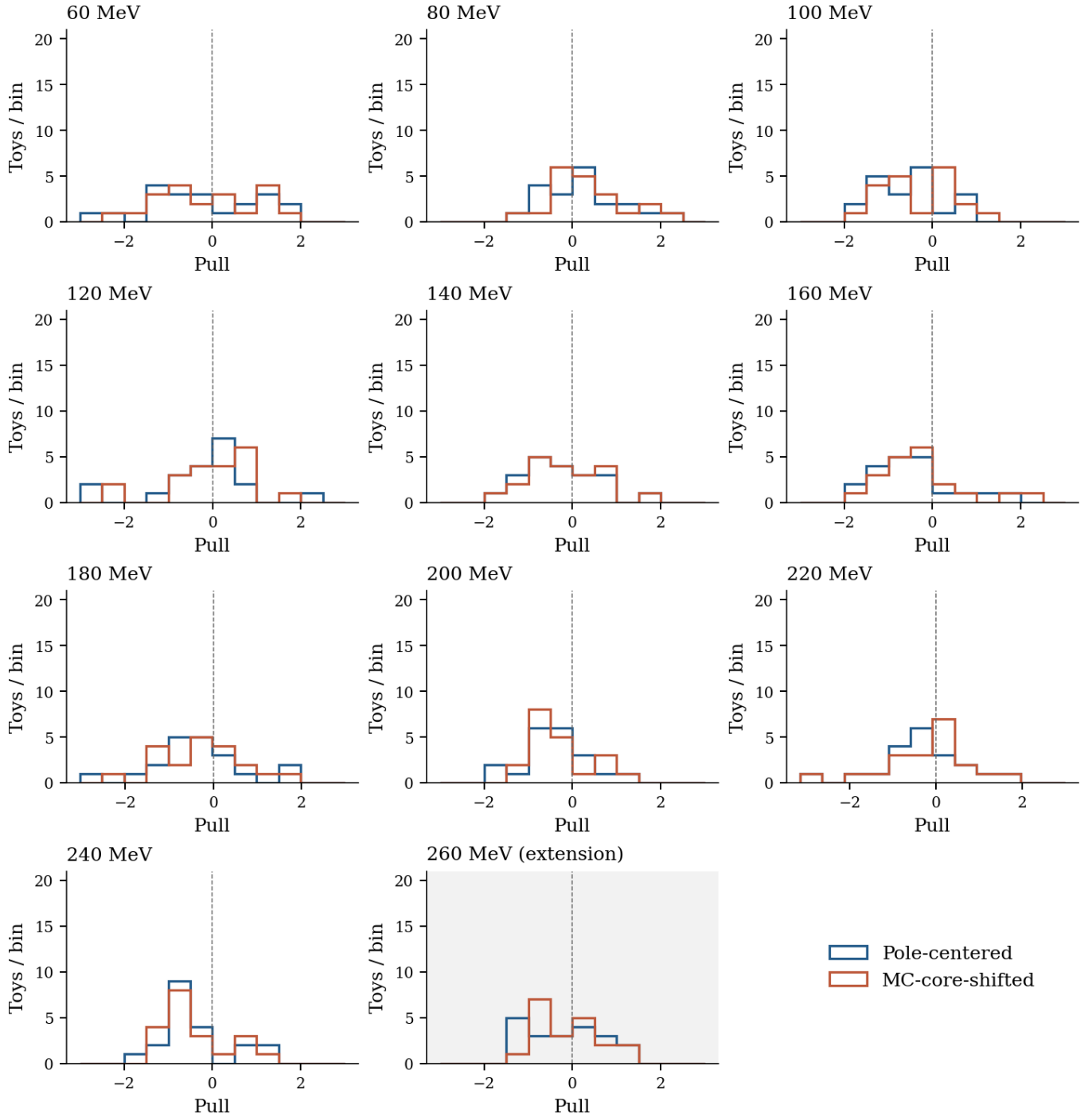
Primary injected-sideband fits. Each outline contains exactly 20 pulls. Both methods share the bin edges within each mass panel; axis ranges may differ between masses. The counts are deliberately not normalized to a smooth density or fitted Gaussian. The shaded 260 MeV panel is the separate extension.

Pull distributions: N = 5,000



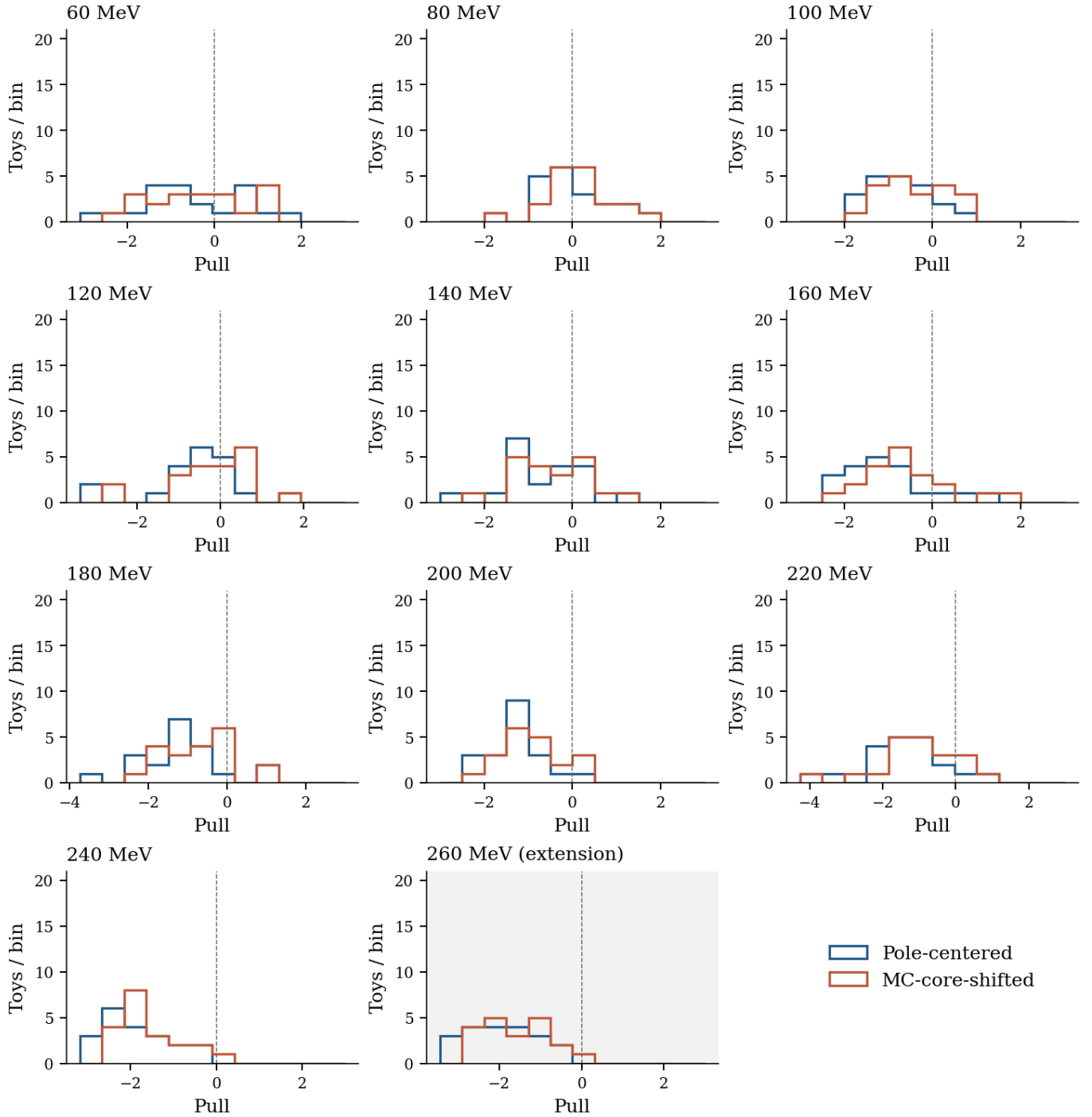
Primary injected-sideband fits. Each outline contains exactly 20 pulls. Both methods share the bin edges within each mass panel; axis ranges may differ between masses. The counts are deliberately not normalized to a smooth density or fitted Gaussian. The shaded 260 MeV panel is the separate extension.

Pull distributions: $N = 10,000$



Primary injected-sideband fits. Each outline contains exactly 20 pulls. Both methods share the bin edges within each mass panel; axis ranges may differ between masses. The counts are deliberately not normalized to a smooth density or fitted Gaussian. The shaded 260 MeV panel is the separate extension.

Pull distributions: $N = 30,000$



Primary injected-sideband fits. Each outline contains exactly 20 pulls. Both methods share the bin edges within each mass panel; axis ranges may differ between masses. The counts are deliberately not normalized to a smooth density or fitted Gaussian. The shaded 260 MeV panel is the separate extension.

Numerical ledger: N = 1,000

All primary injected-sideband cells, including the marked 260 MeV extension. P = pole-centered; C = MC-core-shifted. Mean uncertainty is the average profiled curvature uncertainty. Inclusion columns give toy counts out of 20.

m [MeV]	Mask	Mean A	Mean s	A/N	Pull mean	Pull SD	68%	95%
60	P	-194	19,380	-0.194	-0.06	1.25	9	18
60	C	-41	21,057	-0.041	-0.05	1.11	11	19
80	P	4,386	8,861	4.386	0.38	0.88	14	19
80	C	4,469	9,374	4.469	0.37	0.85	15	19
100	P	-1,162	6,984	-1.162	-0.31	0.75	16	20
100	C	-702	7,360	-0.702	-0.23	0.80	14	20
120	P	1,092	5,708	1.092	0.02	1.11	16	17
120	C	1,280	5,985	1.280	0.05	1.00	17	18
140	P	729	4,761	0.729	-0.06	0.89	15	20
140	C	727	4,942	0.727	-0.06	0.86	17	20
160	P	370	3,950	0.370	-0.16	0.96	14	19
160	C	445	4,147	0.445	-0.13	0.95	15	19
180	P	698	3,412	0.698	-0.09	1.04	15	18
180	C	724	3,549	0.724	-0.08	0.92	15	20
200	P	885	3,041	0.885	-0.04	0.66	17	20
200	C	869	3,132	0.869	-0.04	0.68	17	20
220	P	1,565	2,752	1.565	0.21	1.08	14	18
220	C	1,603	2,833	1.603	0.21	1.08	14	18
240	P	1,648	2,595	1.648	0.25	0.81	16	20
240	C	1,311	2,669	1.311	0.12	0.81	16	20
260 *	P	2,688	2,658	2.688	0.64	0.87	12	18
260 *	C	2,585	2,744	2.585	0.58	0.74	14	18

* 260 MeV extends the original direct-template study and uses the archived 250 MeV kernel state. It is excluded from the overview's 60-240 MeV range summaries.

The 68%/95% columns are conditional profile-likelihood inclusion counts at the injected truth. With 20 trials the fraction changes in steps of 0.05. Exact binomial Clopper-Pearson intervals, mean-bias values, sample errors and all controls are retained in results/summary.csv. These counts are not calibrated detector coverage.

Numerical ledger: N = 5,000

All primary injected-sideband cells, including the marked 260 MeV extension. P = pole-centered; C = MC-core-shifted. Mean uncertainty is the average profiled curvature uncertainty. Inclusion columns give toy counts out of 20.

m [MeV]	Mask	Mean A	Mean s	A/N	Pull mean	Pull SD	68%	95%
60	P	2,624	19,381	0.525	-0.12	1.24	9	17
60	C	2,999	21,058	0.600	-0.09	1.11	11	19
80	P	7,849	8,861	1.570	0.32	0.88	14	19
80	C	8,163	9,375	1.633	0.34	0.85	15	19
100	P	2,366	6,984	0.473	-0.38	0.76	16	20
100	C	3,053	7,361	0.611	-0.26	0.80	14	20
120	P	4,638	5,709	0.928	-0.06	1.11	17	17
120	C	5,050	5,986	1.010	0.01	1.00	17	18
140	P	4,295	4,762	0.859	-0.15	0.90	16	20
140	C	4,469	4,943	0.894	-0.11	0.86	17	20
160	P	3,928	3,951	0.786	-0.27	0.96	14	19
160	C	4,235	4,148	0.847	-0.18	0.94	15	19
180	P	4,186	3,413	0.837	-0.24	1.04	14	19
180	C	4,397	3,550	0.879	-0.17	0.92	14	19
200	P	4,410	3,043	0.882	-0.19	0.65	16	20
200	C	4,536	3,134	0.907	-0.15	0.67	17	20
220	P	4,949	2,754	0.990	-0.02	1.08	15	18
220	C	5,118	2,835	1.024	0.04	1.07	14	19
240	P	4,893	2,598	0.979	-0.04	0.81	14	20
240	C	4,673	2,672	0.935	-0.12	0.80	16	20
260 *	P	5,753	2,663	1.151	0.28	0.88	15	19
260 *	C	5,762	2,749	1.152	0.28	0.75	17	20

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Numerical ledger: N = 10,000

All primary injected-sideband cells, including the marked 260 MeV extension. P = pole-centered; C = MC-core-shifted. Mean uncertainty is the average profiled curvature uncertainty. Inclusion columns give toy counts out of 20.

m [MeV]	Mask	Mean A	Mean s	A/N	Pull mean	Pull SD	68%	95%
60	P	6,311	19,383	0.631	-0.19	1.24	9	18
60	C	7,016	21,060	0.702	-0.14	1.11	10	19
80	P	12,155	8,862	1.216	0.24	0.87	15	19
80	C	12,771	9,376	1.277	0.30	0.84	15	19
100	P	6,843	6,985	0.684	-0.45	0.75	13	20
100	C	7,831	7,362	0.783	-0.29	0.80	14	20
120	P	9,045	5,710	0.905	-0.17	1.11	16	17
120	C	9,743	5,987	0.974	-0.04	1.00	17	18
140	P	8,748	4,764	0.875	-0.26	0.90	15	19
140	C	9,140	4,944	0.914	-0.17	0.86	16	20
160	P	8,271	3,952	0.827	-0.44	0.95	12	20
160	C	8,862	4,149	0.886	-0.27	0.94	14	19
180	P	8,543	3,415	0.854	-0.43	1.05	13	19
180	C	8,995	3,552	0.899	-0.28	0.92	13	19
200	P	8,715	3,045	0.872	-0.42	0.66	16	20
200	C	9,004	3,136	0.900	-0.32	0.68	17	20
220	P	9,146	2,757	0.915	-0.31	1.06	15	18
220	C	9,491	2,838	0.949	-0.18	1.05	15	19
240	P	8,927	2,601	0.893	-0.41	0.80	15	20
240	C	8,852	2,675	0.885	-0.43	0.79	15	20
260 *	P	9,532	2,668	0.953	-0.17	0.87	13	20
260 *	C	9,678	2,755	0.968	-0.12	0.74	17	20

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The 68%/95% columns are conditional profile-likelihood inclusion counts at the injected truth. With 20 trials the fraction changes in steps of 0.05. Exact binomial Clopper-Pearson intervals, mean-bias values, sample errors and all controls are retained in results/summary.csv. These counts are not calibrated detector coverage.

Numerical ledger: N = 30,000

All primary injected-sideband cells, including the marked 260 MeV extension. P = pole-centered; C = MC-core-shifted. Mean uncertainty is the average profiled curvature uncertainty. Inclusion columns give toy counts out of 20.

m [MeV]	Mask	Mean A	Mean s	A/N	Pull mean	Pull SD	68%	95%
60	P	20,551	19,391	0.685	-0.49	1.23	11	18
60	C	22,563	21,068	0.752	-0.35	1.10	11	19
80	P	29,486	8,866	0.983	-0.06	0.88	16	20
80	C	31,258	9,379	1.042	0.13	0.85	16	20
100	P	24,365	6,989	0.812	-0.81	0.75	12	20
100	C	26,502	7,365	0.883	-0.47	0.80	15	20
120	P	26,632	5,714	0.888	-0.59	1.10	15	18
120	C	28,442	5,991	0.948	-0.26	0.99	15	18
140	P	26,667	4,768	0.889	-0.70	0.91	10	19
140	C	27,933	4,949	0.931	-0.42	0.87	13	19
160	P	25,810	3,958	0.860	-1.06	0.95	7	17
160	C	27,575	4,155	0.919	-0.58	0.93	11	19
180	P	26,072	3,422	0.869	-1.15	1.05	8	16
180	C	27,467	3,559	0.916	-0.71	0.93	12	19
200	P	26,083	3,053	0.869	-1.28	0.65	5	17
200	C	27,066	3,144	0.902	-0.93	0.68	10	19
220	P	26,053	2,768	0.868	-1.43	1.09	4	15
220	C	27,046	2,848	0.902	-1.04	1.09	12	17
240	P	25,117	2,616	0.837	-1.87	0.74	4	7
240	C	25,595	2,690	0.853	-1.64	0.75	4	9
260 *	P	24,799	2,691	0.827	-1.93	0.90	3	10
260 *	C	25,584	2,777	0.853	-1.59	0.77	3	13

* 260 MeV extends the original direct-template study and uses the archived 250 MeV kernel state. It is excluded from the overview's 60-240 MeV range summaries.

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